8×8 LED Matrix Mini Fluid Simulation Gadget

User Manual

Device Overview

FLIP Fluid Simulator with a real-time physics simulation displayed on a 64-LED matrix (8×8 grid) at ~43 FPS (23ms per frame). The built-in accelerometer allows the fluid to respond to device tilting.

Button Controls

The device features a single button with three press patterns:

Gesture	Action
Single Press	Cycle through 7 brightness levels (3, 4, 5, 6, 9, 11, 13)
Double-Press	Cycle through 17 color presets (blues, greens, reds, oranges, yellows, purples, pinks, white)
Long Press	Hold 600ms to cycle foam effect threshold (5 levels)

Note: Colors are mixed from RGB (0–255) then compressed to brightness level. Some colors may look identical at low brightness (e.g. at level 3: BLUE1 (0,0,255) → (0,0,3) and PURPLE1 (60,0,255) → (0,0,3)).

Technical Specifications

Component	Specification
Microcontroller	ESP32-S3 @ 240MHz
Display	64 RGB LEDs (8×8 matrix, WS2812B)
Accelerometer	QMI8658 6-axis IMU
Power	220mAh LiPo battery

Open Source Project

Full source code available at:

github.com/vateva/Esp328x8fluid

Built with:

ESP32-S3 | FreeRTOS | C/C++ | FLIP Fluid Simulation

QMI8658 IMU | WS2812B LED Matrix